Liquid biopsy and NGS cancer panels
Tata Memorial: plasma agreed with tissue in 97%

For sales and healthcare teams. Facts summarised in our own words from published sources (links below). Treatment decisions are always the oncologist's.

Indian study: Veldore VH, Choughule A, Routhu T, Mandloi N, Noronha V, Joshi A, Dutt A, Gupta R, Vedam R, Prabhash K. *Validation of liquid biopsy: plasma cell-free DNA testing in clinical management of advanced non-small cell lung cancer.* Lung Cancer: Targets and Therapy 2018;9:1-11. Tata Memorial Centre, Mumbai, with ACTREC and MedGenome. Open access.

Who was tested

- 163 untreated patients with advanced (stage IV) non-small cell lung cancer
- 132 had both tissue and plasma; **31 (about 1 in 5) had no tissue**
- Plasma ctDNA NGS for EGFR exons 19 and 21, compared with tissue PCR

What was found

- Agreement with tissue **97%**; sensitivity 91.1%; specificity 100%
- 4 of 45 tissue EGFR-positive patients were missed in plasma
- Plasma-only group: EGFR found in 16% (abstract figure)
- Authors: useful when tissue is unavailable; not for screening or early diagnosis

Why a broad panel: Indian NGS data

Sharma et al., Frontiers in Genetics 2023 (Hyderabad): 53 lung adenocarcinomas tested on a 46-gene panel. At least one actionable change in **79% (42 of 53)**; EGFR exon 19 deletion or L858R in 20.8%; ALK, RET, NTRK or MET exon 14 fusions in 11.3%. Small, single-centre study of non-metastatic tumours.

IASLC 2021 consensus (Rolfo et al., J Thorac Oncol)

- **At diagnosis:** plasma testing is a valid option and often complements tissue; plasma-first may be acceptable.
- **Progression on targeted medicine:** plasma testing is preferred first; if plasma is uninformative, repeat a tissue biopsy.
- **Monitoring:** an emerging use of plasma testing.
- Use a clinically validated NGS platform; EGFR PCR is an alternative where NGS is not available. A molecular tumour board is advisable.

EPISODE 1 · SALES FIELD SHEET

Liquid biopsy: how to approach doctors

Topic	What to do or say
Doctors to visit	Medical oncologists, pulmonologists, pathologists and cancer hospitals, tumour boards.
Which patients	Advanced lung cancer with too little tissue for gene tests; biopsy not possible or not available; cancer growing again on a targeted medicine; treatment waiting for a gene result.
Opening question	"Doctor, when tissue is not enough for EGFR and ALK, do you send plasma NGS?"
Facts to quote	Tata Memorial: about 1 in 5 patients had no tissue for testing; plasma agreed with tissue in 97%; a positive plasma result was 100% specific.
Test to suggest	Enough tissue: tissue NGS panel. Little or no tissue, or progression: plasma NGS (liquid biopsy). Where NGS is not possible: EGFR PCR first. **Confirm with the lab: genes covered, blood tube, transport, price and report time.**
"Tissue is the gold standard"	"Agreed. Plasma complements it, and a positive plasma result was 100% specific at Tata Memorial."
"What if plasma is negative?"	"Then a tissue test is the next step; a negative plasma result does not rule out a mutation."
Sample	Blood in the special tube the lab specifies; follow the lab's transport time and temperature.
Never say	That it replaces a biopsy for diagnosis, that it screens healthy people, or that a medicine will work. Treatment is the oncologist's decision.

References

- Veldore VH et al. Validation of liquid biopsy: plasma cell-free DNA testing in clinical management of advanced non-small cell lung cancer. Lung Cancer: Targets Ther 2018;9:1-11. dovepress.com
- Sharma R et al. The landscape of actionable genomic alterations in lung adenocarcinomas in India. Front Genet 2023. frontiersin.org
- Rolfo C et al. Liquid biopsy for advanced NSCLC: a consensus statement from the IASLC. J Thorac Oncol 2021. Summary: ascopost.com

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